Andrea Amaduzzi

Research Intern at Niantic Spatial

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Professional Experience

Niantic Spatial — London, UK
Research Intern — Sept 2025 - March 2026

- Designed and fine-tuned multimodal LLMs (10B+ parameters) for spatial reasoning.
- Input modalities used: images, camera poses, depth maps, point clouds, Gaussian Splats.
- Outcome of the project: ECCV 2026 paper submission.
- Advisors: Tommaso Cavallari and Victor Adrian Prisacariu.

Datalogic — Bologna, Italy
R&D Vision Software Engineer — June 2020 - Nov 2021

- Developed and maintained production-grade Python and C++ ML libraries, deployed on millions of edge devices.
- Optimized ML pipelines under real-world constraints, reducing production error rates by >50%.

KUKA–Corporate Research — Augsburg, Germany
Research Intern — July 2019 Jan 2020

- Built deep learning pipelines for semantic segmentation and action recognition using multimodal sensor data.
- Mentored 30+ students on deep learning workflows and 3D sensor projects.

Education

PhD on 3D-Language Understanding — Nov 2021 - March 2026
Computer Vision Laboratory, University of Bologna, Italy
Advisor: Prof. Luigi Di Stefano

- Thesis title: "Multimodal AI for 3D-language Understanding".
- Published 3 first-author publications at top-tier conferences (ICCV, NeurIPS).
- Extensive experience with LLM fine-tuning, prompt design, and large-scale distributed training on LEONARDO CINECA supercomputer.
- Designed custom datasets and benchmarks to measure alignment, grounding, and generation quality.

M.S. in Automation Engineering — Sept 2017 - March 2020
University of Bologna, Italy

- Visiting researcher at the University of Technology of Sydney (GPA: 3.88/4.0 Top 1%).
- Thesis focused on deep learning for human action recognition, at KUKA, Augsburg, Germany.
- Graduation Mark: 110/110 cum laude.
- GPA: 29.23/30.00 (Top 3%).

B.S. in Automation Engineering — Sept 2014 - July 2017
University of Bologna, Italy

- Thesis on sEMG-based Torque Estimation using deep learning.
- Graduation Mark: 110/110 cum laude.
- GPA: 28.71/30.00 (Top 1%).

Publications

Andrea Amaduzzi, Pierluigi Zama Ramirez, Giusepppe Lisanti, Samuele Salti, Luigi Di Stefano
"Spatially-aware Weights Tokenization for NeRF-Language Models", **NeurIPS 2025**

Andrea Amaduzzi, Pierluigi Zama Ramirez, Giusepppe Lisanti, Samuele Salti, Luigi Di Stefano
"Scaling LLaNA: Advancing NeRF-Language Understanding Through Large-Scale Training", **under review at TPAMI**

Andrea Amaduzzi, Pierluigi Zama Ramirez, Giusepppe Lisanti, Samuele Salti, Luigi Di Stefano
"LLaNA: Large Language and NeRF Assistant", **NeurIPS 2024**

Andrea Amaduzzi, Giusepppe Lisanti, Samuele Salti, Luigi Di Stefano
"Looking at words and points with attention: a benchmark for text-to-shape coherence", **ICCVW 2023**

Teaching

Computer Vision and Image Processing M — Fall 24

Logic Design T — Spring 24

Logic Design T — Spring 23

Logic Design T — Spring 22

Skills

Languages:
Italian (mother tongue), English (IELTS 8.0), German (A2), Spanish (A2)

Technologies: Python, Pytorch, Vision Language Models, Large Language Models, Generative Diffusion Models, Distributed training, GitLab, C, C++, LaTeX, Linux, Bash, Javascript.